

SUNCLAVE · DESIGN DECISIONS REGISTER

SunClave design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sunclave		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the TRL 2 and TRL 3 reviews and the design-for-construction work, items to confirm, value engineering and decisions made

SunClave design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Safety: SunClave is a research and educational prototype, not a medical device. Decision 1 below is a pressure-safety decision and must be made before the cooker is ever pressurised with the fittings on its lid.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	How the gauge, relief valve and probe gland are mounted on the cooker	(i) drill the maker's lid with reinforcing washers; (ii) a cooker with factory ports, or an adapter plate on the regulator stem	None: a pressure-safety trade-off for Amish. The calculation (SCL-CAL-001 section 8) applies to both	Build plan step 19 (lid and sensors) and safety stop S6; nothing is pressurised until this is decided	SCL-DDR-001, item 5
2	Accept the design-for-construction changes C1 to C9	Accept; accept with changes; reject	Accept. They keep what SunClave does and make every part makeable and every joint bolted. They also settle the four appearance-model differences noted on 2026-09-26 (logger box height, transducer on the tee, cooker handles, petal seams)	The whole build plan	SCL-DDR-003, Table 1
3	R16 mass margin of 0.6 kg (44.4 kg against 45 kg)	(a) accept, weigh the prototype at TRL 4; (b) take about 0.9 kg out now (5 mm yoke plates, 20 x 20 mm holder arms)	(a), with (b) ready	Yoke plates (section 3.14) and holder arms (section 3.15) if (b)	SCL-DDR-003, A1
4	Tilt lock: friction only, or a back-up pin	(a) two friction locks as modelled; (b) add a drop pin through holes in the fan	(a) for the prototype; check the slip torque at TRL 4	Yoke plates and steps 13 and 14 if (b)	SCL-DDR-003, A2
5	First partner and site for co-design	Chosen per area later (portfolio rule)	None yet	Not part of the build; sets the altitude and the cooker model	SCL-DDR-001, item 12

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The cooker's maker's rating for 103.4 kPa, its outside diameter (about 288 mm), and handles whose undersides are about 28 mm below the rim and reach at least 214 mm from the centre	The rating is R8; the handles must rest on the 408 mm holder ring and the jacket must clear its 348 mm hole	SCL-DDR-001 item 4, SCL-DDR-003 C3
2	The reflector sheet takes a 12 mm flange fold without cracking its bright layer	The petals are fixed by riveting through these flanges	SCL-DDR-003 C6
3	A 20 x 3 mm bar bends on edge in the plywood jig without buckling (bend one rib first)	If it buckles, the ribs need a heavier bar or a fabricator's rolls	SCL-DDR-003 C6
4	Castor mounting height 102 mm and a top-plate hole pattern on a 42 mm square, all four with brakes	Sets the stand height and the drilling of the cross rails	SCL-DDR-003 C5
5	The power bank has a low-current (always-on) mode	The logger draws about 72 mA; many banks switch off below 50 to 100 mA	SCL-CAL-001 section 11
6	The relief valve's seat is 4 mm or more and its set pressure is 125 kPa gauge or less, preferably certified	R8 relief capacity	SCL-CAL-001 section 8
7	The pressure transducer's overpressure rating is 600 kPa or more	It must survive a relief lift	SCL-DDR-002 item 18

■ Value engineering

Value-engineering target: USD 450 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 497 (USD 47 over the target), validation consumables (USD 45) excluded. Main cost drivers and savings worth trying:

- The largest lines are the reflector petals (USD 75), the cycle logger (USD 66), the timber stand with castors, axles and plates (USD 58), the cooker (USD 55) and the lid gland, probe and tee (USD 30).
- Making the design constructable added USD 57 (from USD 440): the yoke plates, stand-offs and locks (line 3, USD 18 to 30), the stand's axle plates, axles, collars and brackets (line 4, USD 44 to 58), the clips (line 2), the plate ring and brackets (line 5), the larger logger box and plug-in sensor leads (line 14) and more bolts (line 17).
- Savings worth trying: have one local cutting shop cut the yoke plates, axle plates, holder ring and hub plate from one sheet of steel (likely cheaper than four separate jobs); price aluminised film on plain 0.5 mm aluminium against bright reflector sheet; buy the logger modules as one combined board; buy bolts in boxes of 50 rather than by the piece.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 4 and 6 to 11: altitude option B, budget cuts with USD 450, pitch wording, 12 L household cooker, level cooker at the focus, petal dish, gnomon aiming, load limits, logger pass criteria, backup heat	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	SCL-DDR-001
2026-09-25	Items 13 to 18: retarget every 12 min, 45 kg total, whole vessel a marked hot zone, four locking castors, trimming rule, 0 to 300 kPa transducer	Amish: "i accept all your recommendations, go with them across all repos."	SCL-DDR-002
2026-09-25	TRL 4 is on hold for the whole portfolio	Amish: "Make sure we don't proceed to TRL 4 on any of them"	SCL-DDR-001
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SCL-DDR-003 (draft, open for review as decision 2)
2026-09-30	Open decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	This register
2026-10-01	budget_usd is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register, Value engineering